Minimum Wages, Efficiency, and Welfare

Adam Suraj

Based on Berger, Herkenhoff, and Mongey (2025)

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Research Question: Can a Minimum Wage Raise Efficiency?

The problem

- In minimum-wage debates, proponents often invoke **monopsony**: employers have wage-setting power and hire too few workers.
- If that mechanism is important, a moderate wage floor can raise both wages and employment rather than destroy jobs.
- But policy requires a number: **how high should the floor be if the objective is aggregate efficiency?**

$$w_i = \mu_i \, \mathrm{MRPL}_i, \qquad \mu_i < 1$$

$$\Downarrow$$

$$n_i < n_i^{\text{efficient}}$$

- Labor markets are geographically distinct.
- Firms differ in productivity, employment, wages, and markdowns.
- One common minimum wage acts on all of them at once.

Source: Berger, Herkenhoff, and Mongey (2025), Introduction and Section 3.3.

Theoretical Model I: Markets and Heterogeneous Firms

Labor markets

- A continuum of geographically distinct markets $j \in [0,1]$; market j contains M_j firms.
- Jobs are closer substitutes within a market than across markets:
$$n_j = \mathrm{CES}_\eta(\{n_{ij}\}_i),$$
$$N = \mathrm{CES}_\theta(\{n_j\}_j), \qquad \eta > \theta.$$
- η: substitution between firms in the same market; θ: substitution across markets.
- Market size and the number of competing firms vary, generating differences in concentration.

Heterogeneous firms / entrepreneurs

Each firm has permanent productivity z_{ij} and produces
$$y_{ij} = Zz_{ij}\left(n_{ij}^{\gamma} k_{ij}^{1-\gamma}\right)^{\alpha}.$$

- $\alpha < 1$ gives decreasing returns; γ is labor's share inside the capital-labor composite.
- Firms choose $(w_{ij}, n_{ij}, k_{ij}, \bar{n}_{ij})$ to maximize
$$\pi_{ij} = y_{ij} - Rk_{ij} - w_{ij}n_{ij},$$

subject to $w_{ij} \geq \underline{w}$, $n_{ij} \leq \bar{n}_{ij}$, and their residual labor supply. They compete Cournot-style, taking rivals' employment as given.

Source: Berger, Herkenhoff, and Mongey (2025), Section 2 and Section 4.3.1.

Theoretical Model II: Workers Create Firm-Level Labor Supply

Household choice

A representative household chooses consumption, saving, and employment at every firm:

$$\max_{C,K',\{n_{ij}\}} U(C,N)$$

subject to its budget and any firm-level hiring caps. Labor is a nested CES aggregate:

$$n_j = \left[\sum_{i=1}^{M_j} n_{ij}^{(\eta+1)/\eta}\right]^{\eta/(\eta+1)},$$

$$N = \left[\int_0^1 n_j^{(\theta+1)/\theta}\, dj\right]^{\theta/(\theta+1)}, \qquad \eta > \theta.$$

Quick Summary of Important Staff

- Each firm faces an upward-sloping residual labor-supply curve.
- Hiring more workers requires a higher wage. Firms internalize this when choosing employment.

Employment is below the efficient level, where the worker's marginal cost of labor equals marginal revenue product.

$$w_{ij} = \mu_{ij}\,\mathrm{MRPL}_{ij}, \qquad 0 < \mu_{ij} < 1.$$

Homogeneous workers make welfare changes pure efficiency changes; redistribution is deliberately shut down.

Source: Berger, Herkenhoff, and Mongey (2025), Sections 2 and 3.3.

What Rationing Means in the Model

Why a hiring cap is needed

- A binding wage floor need not clear labor supply at each firm.
- At the posted wage, workers may want more jobs than the firm finds profitable to provide.
- The firm posts a cap $\bar{n}_i$: the maximum number of workers it will hire.

$$n_i \leq \bar{n}_i, \qquad \mathrm{MRPL}_i(\bar{n}_i) = \underline{w}.$$

Numerical example

Suppose $\underline{w} = \$15$.

- At \$15, workers would supply $n_i^s = 120$ jobs to the firm.
- The firm wants only $n_i^d = 80$, because the 80th worker has $\mathrm{MRPL}_i = \$15$.
- It sets $\bar{n}_i = 80$ and hires 80 workers.
- The remaining 40 do not queue at this firm: they observe the cap and choose other firms or markets.

Source: Berger, Herkenhoff, and Mongey (2025), Section 2 and equation (9).

Shadow Wages, Markdowns, and Markups

Definitions inside the model

$$\widetilde{w}_i = p_i w_i = \mathrm{MRS}_i(n_i), \qquad 0 < p_i \leq 1.$$

$$\widetilde{\mu}_i = \frac{\widetilde{w}_i}{\mathrm{MRPL}_i} \quad \text{(shadow markdown)},$$

$$\widetilde{m}_i = \frac{\mathrm{MRPL}_i}{\widetilde{w}_i} = \frac{1}{\widetilde{\mu}_i} \quad \text{(shadow markup)}.$$

- **Shadow wage $\widetilde{w}_i$:** the allocative wage consistent with observed employment.
- $p_i = 1$ without rationing; $p_i < 1$ when posted pay exceeds that allocative wage.

Continue the $15 example

The firm hires $n_i = 80$ workers at posted pay $w_i = \$15$. At this quantity, inverse labor supply gives $\mathrm{MRS}_i = \$12$, while $\mathrm{MRPL}_i = \$15$.

$$p_i = \frac{12}{15} = 0.8, \qquad \widetilde{w}_i = p_i w_i = \$12,$$

$$\widetilde{\mu}_i = \frac{12}{15} = 0.8, \qquad \widetilde{m}_i = \frac{15}{12} = 1.25.$$

Meaning: posted pay equals MRPL, yet a 20% shadow markdown remains because posted pay does not clear firm-level labor supply.

Source: Berger, Herkenhoff, and Mongey (2025), Section 3.2. Numerical values are illustrative.

How a Minimum Wage Can Raise Efficiency

Channel	Equilibrium response	Efficiency logic
1. Direct	The floor binds at a low-wage firm; its wage and employment can rise before rationing begins.	The firm's shadow markdown $\widetilde{\mu}_i$ moves toward one.
2. Spillover	As the constrained firm becomes more attractive, unconstrained competitors respond by raising wages.	Competitors' shadow markdowns narrow even though the floor does not bind there.
3. Reallocation	Jobs lost at rationed low-productivity firms are reallocated mainly within the local market.	Employment can move toward higher-productivity firms, improving allocative efficiency ω.

Quantitative tension

The same floor can also create rationing and move labor down the productivity ladder. The model must aggregate the gains and losses across all firms and markets.

Source: Berger, Herkenhoff, and Mongey (2025), Introduction, Sections 3.3–3.5.

Calibration to the U.S. Economy

Data and empirical market definition

- 2014 Longitudinal Business Database.
- 2019 Current Population Survey.
- BLS and national accounts.
- Empirical market: 3-digit NAICS industry $\times$ commuting zone.
- Solution uses $J = 5{,}000$ markets.

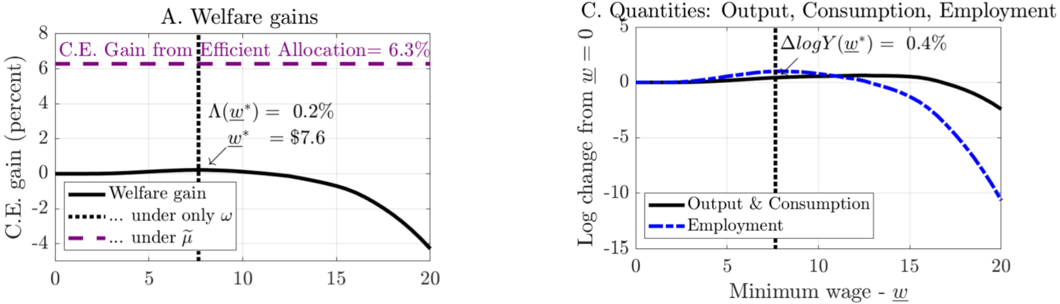
Object	**Value**	**Role in the model**
η	10.85	Substitution across firms within a market
θ	0.42	Substitution across geographically distinct markets
$\sigma_{\log z}$	0.31	Dispersion in permanent firm productivity
α	0.94	Decreasing-returns curvature in production
Mean M_j	113.1	Number of firms per empirical market
HHI	0.11	Payroll-weighted market concentration
$w <$ \$15	30.6%	Share of workers initially below the policy threshold

What these values imply

$\eta \gg \theta$ makes local substitution and reallocation important. Productivity dispersion generates large differences in firm size and markdowns. With α close to one, marginal revenue product declines slowly, so employment can contract sharply under rationing.

Source: Berger, Herkenhoff, and Mongey (2025), Section 4.1 and Table I.

Main Quantitative Result



- At the efficiency-maximizing floor of about \$7.65, welfare rises by 0.22% and output and consumption by 0.40%.
- Eliminating monopsony power yields 6.3% welfare. A minimum wage captures only about 3% of that potential.

Source: Berger, Herkenhoff, and Mongey (2025), Figure 4A, Figure 4C, and Section 4.2.

Why the Gains Are Small: Corner Store vs. Supermarket

	Corner store	**Supermarket**
Productivity and size	Low productivity, small employment share	High productivity, large employment share
Initial monopsony wedge	Faces elastic labor supply; markdown is already narrow	Faces less elastic labor supply; markdown is wide and accounts for much of the aggregate distortion
Low minimum wage	Binds early and can raise employment, but the efficiency gain is small	Usually does not bind; responds little to a small competitor's wage increase
High minimum wage	Quickly becomes too high and creates severe hiring rationing	Finally begins to narrow the large distortion

The targeting problem

A common floor cannot reach the supermarket's large monopsony wedge without first over-correcting the corner store.

Source: Berger, Herkenhoff, and Mongey (2025), Section 4.3.1 and Figure 5.

Decomposing the Small Positive Gain

Monopsony creates a large efficiency problem, but a minimum wage captures only a small part of the available solution.

Component	Share at $\underline{w}^*$	Economic meaning
Narrower shadow markdowns $\widetilde{\mu}$	50%	Mainly wage spillovers: unconstrained competitors raise wages and narrow their wedges.
Better allocation ω	50%	Reallocation: workers move from rationed low-productivity firms toward more productive local employers.

Why this is not primarily the textbook direct effect

- Baseline granular oligopsony with strategic interaction: $\underline{w}^* = \$7.65$.
- With effectively infinitesimal firms $(\eta = \theta)$, spillovers and local reallocation disappear: $\underline{w}^* = \$0.70$. Direct effects alone are nearly zero.

Source: Berger, Herkenhoff, and Mongey (2025), Figure 4B, Section 4.3.2, and Table II.

Policy Takeaway

What the standard monopsony argument gets right

- A wage-setting employer pays below marginal revenue product and hires too little.
- A moderate floor can raise wages and employment simultaneously.
- Employment losses are therefore not an automatic consequence of every minimum-wage increase.

Why a uniform floor is ineffective as efficiency policy

- It reaches small, relatively competitive firms first.
- Firms with the largest wedges often pay higher wages and barely respond.
- Raising the floor far enough to reach them creates rationing elsewhere.

Bottom line

Positive employment effects are consistent with monopsony, but they do not imply that a minimum wage is an effective instrument for removing aggregate monopsony distortions.

Source: Berger, Herkenhoff, and Mongey (2025), Sections 4.2–4.3 and Conclusion.

What I Value Most

The paper models monopsony seriously—and follows that choice all the way through

1. Once a minimum wage binds, the competitive model predicts higher labor supply and lower labor demand, creating an excess-supply gap that does not have any economic meaning and is empirically false.
2. Paper explicitly models wage-setting firms, hiring limits, and worker reallocation. Rationing becomes part of equilibrium rather than an unexplained gap and generally helps us get some helpful intuition about how minimum wage works.

Source: Berger, Herkenhoff, and Mongey (2025), Sections 2–3 and Conclusion; presenter's interpretation.